

Akila Abenayaka

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[GitHub](#) | [LinkedIn](#) | [Medium](#) | [Portfolio](#)

SOFTWARE ENGINEERING UNDERGRADUATE

Passionate Software Engineering undergraduate at the University of Kelaniya with strong interests in **Generative AI**, **Full-Stack Development**, and **Cloud-Native Systems**. Experienced in developing scalable web applications using MERN and Spring Boot ecosystems while exploring AI-powered solutions using LLMs and RAG architectures. A fast learner and collaborative team player committed to building impactful, intelligent software solutions.

EDUCATION

BSc(Hons) Software Engineering
University of Kelaniya

2022–Present
Sri Lanka

- Specializing in **Net-Centric Web Application Development**, **Data Science & Engineering**, and **Mobile Computing**
- CGPA: **3.729/4.0**

GCE Advanced Level
Kegalu Vidyalaya

2007–2020
Kegalle, Sri Lanka

- Completed course work in **Combined Mathematics**, **Physics**, and **Chemistry**
- 2A & 1C Passes** | Z-Score: **1.589**

PROJECTS

AI-Driven Intelligent Chatbot | Individual Project

Python, Gemini 1.5 Pro, PostgreSQL

[GitHub](#)

- Developed an AI-powered chatbot utilizing **Gemini 2.5 Pro** and **Retrieval-Augmented Generation (RAG)** for context-aware responses
- Built a custom knowledge retrieval system using **PostgreSQL** for efficient document querying and low-latency responses
- Integrated vector search and PDF querying capabilities to enhance external knowledge access
- Optimized prompt engineering techniques to improve conversational quality and response accuracy

Rabbit E-Commerce Platform | Individual Project

MERN Stack, Redux Toolkit, Stripe

[GitHub](#)

- Engineered a scalable full-stack clothing marketplace with secure authentication and responsive UI design
- Managed application-wide state using **Redux Toolkit** to ensure seamless frontend dataflow
- Integrated **Stripe** for secure payment processing and **Cloudinary** for media management
- Implemented RESTful API and optimized frontend responsiveness across mobile and desktop devices

ReserveX Vendor Management System | Group Project

Spring Boot, React, PostgreSQL

[GitHub](#)

- Developed a vendor reservation and management system for BMICH bookfair operations
- Built secure backend services using **Spring Boot**, **JWT**, and **Spring Security**
- Designed scalable architecture patterns to improve maintainability and booking performance
- Created professional dashboard for vendor and inventory management workflows

- Contributed to the official TEDxUOK event website with emphasis on UI/UX and mobile responsiveness
- Integrated **Supabase** services for secure event registration and real-time backend functionality
- Improved website SEO and optimized performance for faster asset loading and user engagement

TECHNICAL SKILLS

Languages	: Java, JavaScript, TypeScript, Python, C, HTML, CSS, SQL
Frontend	: React.js, Next.js, Redux Toolkit, Tailwind CSS, Bootstrap
Backend	: Node.js, Express.js, Spring Boot, REST APIs
AI & Data	: Generative AI, Gemini API, RAG Architecture, Prompt Engineering
Databases	: PostgreSQL, MongoDB, MySQL, Redis
Tools & Platforms	: Git, Docker, Firebase, Supabase, Postman, Vercel

CERTIFICATIONS & ACHIEVEMENTS

- **First Place** – Paramount Math IQ Competition, Department of Mathematics, University of Kelaniya
- **Generative AI Fundamentals** – Google Cloud Training (2024)
- **Java (Basic/Intermediate) Certification** – HackerRank (2023)

VOLUNTEER EXPERIENCE & LEADERSHIP

Head of Logistics Software Engineering Students' Association (SESA)	2025 – Present University of Kelaniya
• Coordinating logistics and operational planning for technical events and workshops • Managing resource allocation and vendor coordination for student activities	
Digital Marketing Lead IEEE Student Branch	2024 – 2026 University of Kelaniya
• Leading digital outreach campaigns and social media engagement for IEEE events • Designed promotional content and improved audience reach for technical webinars	

LANGUAGES

English	: Professional Proficiency
Sinhala	: Native Proficiency

REFERENCES

• Dr. Nalin Warnajith Senior Lecturer (Grade II), Software Engineering Teaching Unit, University of Kelaniya nwarnajith@gmail.com +94 71 7570140	• Dr. Isuru Hewapathirana Head, Software Engineering Teaching Unit, University of Kelaniya ihewapirana@kln.ac.lk +94 777 329626
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